

Ice Core (grade 6 edition) — play-through review

Theme `icecore_ms` · middle-school earth science, grade 6 · 10 days, 33 stops · reviewed 2026-08-21 by reading `books/icecore-ms.yml` in full, working every board, and comparing against the parent `icecore`.

Verdict

The junior edition with the most arithmetic done *in words*, and it works. Eight boards, and the solutions are written the way a sixth grader would say them aloud — "*Minus fifty-four divided by thirty is minus 1.8 degrees*", "*Eighty-eight ÷ 0.32 is about 275 years*", "*Sixteen ÷ twelve is about 1.3 years*". All eight are correct:

Board	Arithmetic	Idea
How thick a year is here	$0.32 \times 340 \div 917 \approx 0.12 \text{ m}$	snow to ice equivalent
One year proves nothing	$-54 \div 30 = -1.8 \text{ }^{\circ}\text{C}$	an average of thirty readings
How old the bottom of the snow is	$88 \div 0.32 \approx 275 \text{ years}$	a depth divided by a rate
What the ratio says in degrees	$1.8 \div 0.67 \approx 2.7 \text{ }^{\circ}\text{C}$	a proxy turned into a temperature
How much younger the air is	$88 \times 0.62 \div 0.119 \approx 460 \text{ years}$	the gas-ice age difference
How much more carbon dioxide	$90 \div 190 \times 100 \approx 47\%$	a percentage increase
The clock that does not need layers	$5,730 \times 2 = 11,460 \text{ years}$	two half-lives
The shortest thing this ice can show	$16 \div 12 \approx 1.3 \text{ years}$	resolution as a ratio

The gas-ice age difference survives the derivation, which is the surprising part: the parent's whole resolution is that two records were compared on two different clocks because a bubble and the ice around it are 458 years apart, and the junior edition computes 460 with one multiplication and one division. That is a genuinely subtle finding delivered at grade 6 without softening it.

Answerable: 33/33. **Sense:** Strong. The disagreement with a camp four hundred kilometres away drives all ten days. **Level:** Right. `questionLoad` reports 3 of 33 demanding stops (9%). **Fun:** Good. A plane in fifteen days and "anything not measured by then waits two years" is a clean deadline.

Implemented since this review

- **IM-01** was fixed during the review.
- **IM-02** `dayNoun`.
- Numeral damage.

Everything else in the findings table below is unchanged, and `npm run check` is green on this theme.

Findings

ID	Sev	Where	Issue	Proposed fix
IM-01	FIXED	Two passages	The brittle-zone depth range had been destroyed by numeral normalisation — " <i>Ice between 51200 metres holds bubbles at pressures the surface cannot support</i> ", where "500 and 1," had been swallowed. Repaired to "between 500 and 1,200 metres" earlier in this review, in this book and twice in the parent. Re-imported, both themes green.	Done.
IM-02	WORTH	themes/icecore_ms/theme.js	No dayNoun . The plan card prints "Day N" while the campaign's clock is "the plane comes in fifteen days" — and the campaign is ten days long, so the plan card's number and the aircraft's number are two different counts with no help telling them apart. This is the most confusing instance of the finding in the junior set.	Count down to the plane. Cross-campaign §4.
IM-03	WORTH	Days 2, 3 and 6	Three of ten days author 4 stops, so those three carry no callback.	Prefer a move to a delete: days 7–10 all carry three.
IM-04	WORTH	Zero — Review variants	No spaced retrieval. Cross-campaign §9.	Three or four variants.
IM-05	WORTH	engine/dev/concept-debt.json , 7 rows	Seven ordering rows, no takesAsRead at grade 6. Two are within-day pairs — <i>a ratio can be turned into a temperature resting on something measured today can stand in for something nobody measured then, and averages, and why one year proves nothing</i> resting on graph reading — and a prerequisite beside its dependent is met second by half the players.	Read the two same-day pairs first; a move is cheaper than a written claim.
IM-06	WORTH	engine/dev/curriculum-debt.json , 2 rows	speed = distance ÷ time and time = distance ÷ speed recorded as uncomputed. Same prose-equation matcher caveat as blackout_ms 's BM-01: check the boards before treating these as content work. "How old the bottom of the snow is" computes $88 \div 0.32$, which is a depth divided by a rate — arguably the second of the two.	Re-check, then write whichever is genuinely absent.
IM-07	TASTE	Numeral damage, 2 occurrences	"1 of", "1 cm". The parent has eleven, including the two that were fixed.	Editorial pass.

What the derivation did well

- **It kept the gas-ice age difference.** The hardest idea in the parent, computed at grade 6, with the right answer.
- **It kept the disagreement as the plot.** "A disagreement kept quiet is a decision made for everybody else" is Marit Halvorsen's line on the opening card and it is a better statement of scientific ethics than most of the senior campaigns attempt.
- **The prose is written to be read aloud.** "Minus fifty-four divided by thirty is minus 1.8 degrees" is how the arithmetic should look at this level, and eight boards do it consistently.

Opening and closing

Opening: 111 words, names Marit Halvorsen with the job attached and gives her the campaign's ethical position. The best-argued junior opening in the set.

Closing: three paragraphs, inheriting the parent's hedged-sentence ending — "the report said so in the hedged sentence rather than the confident one" — which is a remarkable thing to keep at grade 6 and the right thing to keep.

Warm-ups

Authored for this edition. No findings.